Atomic Bonding in Solids

Interatomic Bonding

- The arrangement of atoms in a solid is mainly influenced by the nature of the chemical bond that holds them together. The atoms and molecules in a solid state are more closely packed than in the gaseous and liquid states and are held together by strong mutual forces of attraction and repulsion.
- A molecule is composed of atoms by the interatomic bonding, and then the solid is composed of atoms or molecules. The interatomic bonding can be divided into two types: the chemical bond and the physical bond.
- Bonds hold atoms at different distances such that the inter-atomic forces just balance. The process of holding is known as bonding.
- The attractive forces between atoms are basically electrostatic in origin and the classification of different types of bonding depends on the electronic structure of the atoms concerned and are hence directly related to the periodic table.
- The type of bonding determines the electrical, chemical and physical properties of the material concerned.

Bonding Forces and Energies

- Considering the interaction between two isolated atoms as they are brought into close proximity from an infinite separation.
- At larger distances, the interactions are negligible.
- As the atoms approach, each exerts forces on the other.
  - Attractive (F_A)
  - Repulsive (F_R),
  - Separation or interatomic distance (r).
- Ultimately, the outer electron shells of the two atoms begin to overlap, and a strong repulsive force comes into play.

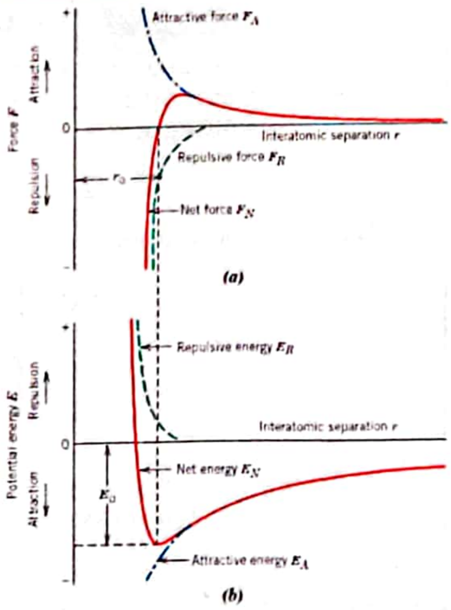


Fig. (a) The dependence of repulsive, attractive, and net forces on interatomic separation for two isolated atoms. (b) The dependence of repulsive, attractive, and net potential energies on interatomic separation for two isolated atoms.

Bonding Forces and Energies

- Figure (a) shows the schematic plot of F_A and F_R versus r. The origin of an attractive force F_A depends on the particular type of bonding that exists between the two atoms.

- Repulsive forces arise from interactions between the negatively charged electron clouds for the two atoms and are important only at small values of r as the outer electron shells of the two atoms begin to overlap. (Figure (a))

- The net force F_N *between the two atoms is just the sum of both attractive and* repulsive components; that is,

$$\boxed{F_N = F_A + F_R \quad \text{.......................1}}$$

When F_A *and* F_R *are equal in magnitude but opposite in sign, there is no net force*— that is,

$$\boxed{F_A + F_R = 0 \quad \text{..........................2}}$$

- The centers of the two atoms remain separated by the equilibrium spacing r_0, *as indicated in Fig. (a) For many atoms,* r_0 *is approximately* 0.3 nm.

Bonding Forces and Energies

- Sometimes it is more convenient to work with the potential energies between two atoms instead of forces. Mathematically, energy *(E) and force (F) are related as*

$$\boxed{E = \int F\,dr \quad \text{.................................3}}$$

And, for atomic systems,

$$E_N = \int_r^{\infty} F_N\,dr \quad \text{..............................4}$$

$$= \int_r^{\infty} F_A\,dr + \int_r^{\infty} F_R\,dr \quad \text{..............................5}$$

$$= E_A + E_R \quad \text{..............................6}$$

Where, E_N, E_A, *and* E_R *are the net, attractive, and repulsive energies, respectively, for* two isolated and adjacent atoms

Bonding Forces and Energies

- Figure b plots attractive, repulsive, and net potential energies as a function of interatomic separation for two atoms. From Equation 6, the net curve is the sum of the attractive and repulsive curves.
- The minimum in the net energy curve corresponds to the equilibrium spacing, r_0. Furthermore, the bonding energy for these two atoms, E_0, corresponds to the energy at this minimum point (also shown in Figure b); it represents the energy required to separate these two atoms to an infinite separation

Force in Equation 3 may also be expressed as

$$F = \frac{dE}{dr} \quad \text{..............................7}$$

Likewise, the force equivalent of Equation 2.8a is as follows:

$$F_N = F_A + F_R \quad \text{..............................8}$$

$$= \frac{dE_A}{dr} + \frac{dE_R}{dr} \quad \text{..............................9}$$

- For example, materials having large bonding energies typically also have high melting temperatures; at room temperature, solid substances are formed for large bonding energies, whereas for small energies, the gaseous state is favored; liquids prevail when the energies are of intermediate magnitude.

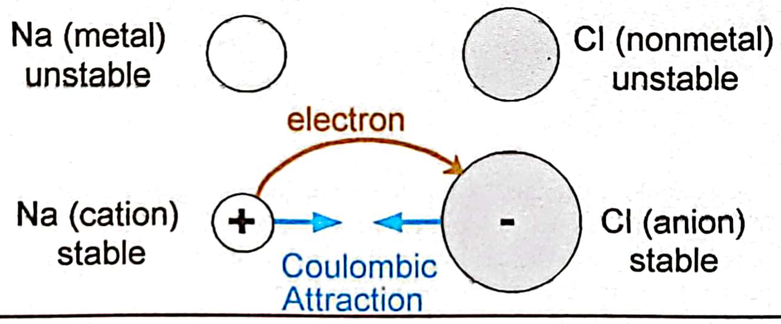
Primary Interatomic Bonds

- Primary bonds are the strongest bonds between atoms by virtue of their interatomic nature. These bonds are also known as attractive bonds.
- The attractive forces are directly associated with the valence electrons in the respective orbitals. Primary bonds are further classified on the positions taken by bond electrons during the formation of the bond.

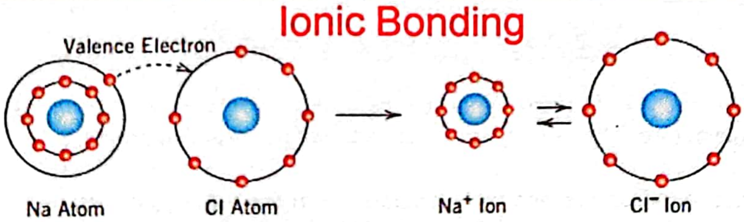
The principal types are :
(i) Ionic or electrostatic bonds
(ii) Covalent, atomic or homopolar bonds
(iii) Metallic bonds

Ionic Bonding

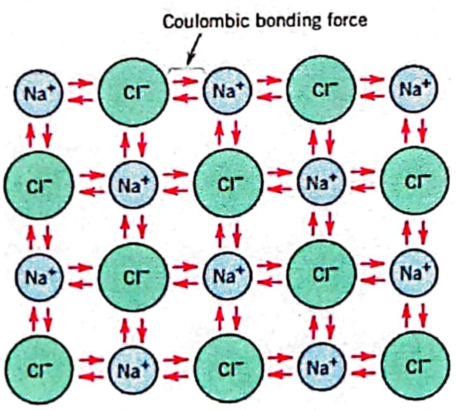
- ❑ An ionic bond can only be formed between two different atoms, one electro- positive and the other electronegative. These positive and negative ions, when brought into close proximity, result in an attractive force which forms the ionic bond.
  - Occurs between + and - ions.
  - Requires electron transfer.
  - Large difference in electronegativity required.
  - Example: NaCl



Ionic Bonding



(a)

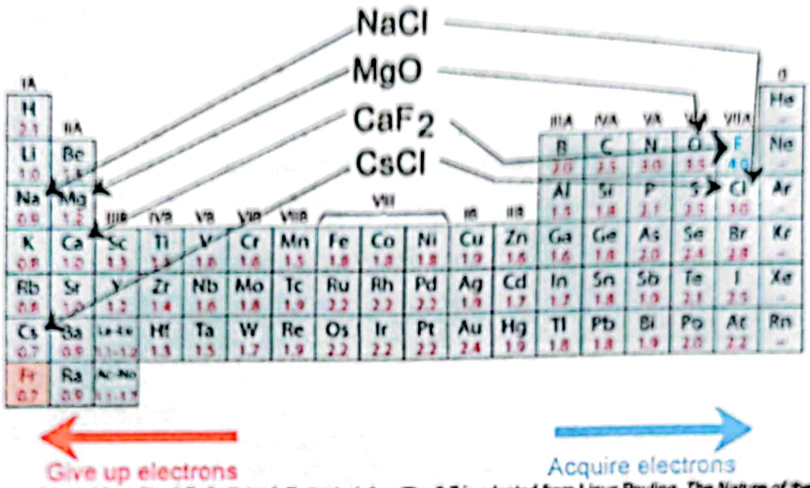


(b)

Figure 2.11 Schematic representations of (*a*) the formation of Na+ and Cl- ions and (*b*) ionic bonding in sodium chloride (NaCl)

Examples: Ionic Bonding

- Predominant bonding in Ceramics



Adapted from Fig. 2.7, *Callister & Rethwisch 8e.* (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

Characteristics of Ionic Compounds

- In ionic bonding, a metallic element loses from the outer electron shell of its atom a number of electrons equal to its numerical valency. Thus, an electrostatic attraction between positive and negative ions occurs.
- Ionic compounds are generally crystalline in nature and rigid.
- They have high melting and boiling points due to the strong electro-static forces binding them.
- Since each positive ion attracts all neighboring negative ions and vice versa, the bond itself is non-directional.
- They are generally non-conductors of electricity.
- They are insoluble in organic solvents but highly soluble in water.

EXAMPLE PROBLEM 2.2

Computation of Attractive and Repulsive Forces between Two Ions

The atomic radii of K^+ and Br^- ions are 0.138 and 0.196 nm, respectively.

(a) Calculate the force of attraction between these two ions at their equilibrium interionic separation (i.e., when the ions just touch one another).

(b) What is the force of repulsion at this same separation distance?

Covalent Bonding

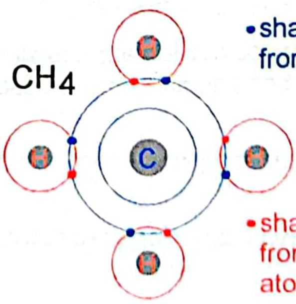
The covalent bond is formed by sharing of electrons between atoms rather than by transfer of electrons. Only a few solids are held together by covalent bonds.

- similar electronegativity ∴ share electrons
- bonds determined by valence – *s* & *p* orbitals dominate bonding
- Example: CH_4

C: has 4 valence e^-,
needs 4 more

H: has 1 valence e^-,
needs 1 more

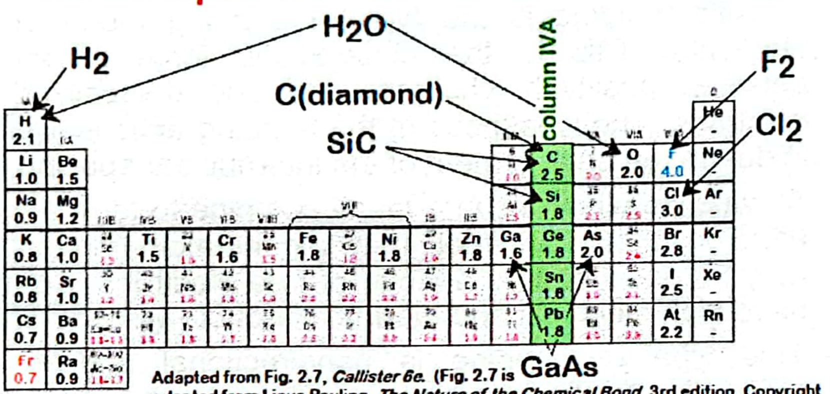
Electronegativities
are comparable.



• shared electrons from carbon atom

• shared electrons from hydrogen atoms

Examples: Covalent Bonding



Adapted from Fig. 2.7, *Callister 6e.* (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University. From Callister 6e resource CD.

- Molecules with nonmetals
- Molecules with metals and nonmetals
- Elemental solids (RHS of Periodic Table)
- Compound solids (about column IVA)

Characteristics of Covalent Compounds

(i) Covalent compounds are directional in nature.

(ii) They can exist in all states of matter.

(iii) They are generally electric insulators.

(iv) They are homopolar, i.c., the valence electrons are bound to individual or pairs of atoms and electrons cannot move freely through the material as in the case of metallic bonds.

(v) They are insoluble in water, but soluble in non-polar solvents such as benzene, alcohol, chloroform and paraffins.

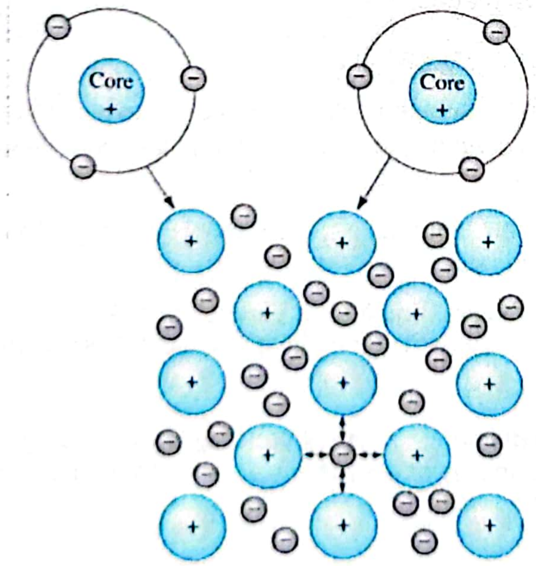
(vi) They are soft, rubbery elastomers, and form a variety of structural materials commonly known as plastics. Their melting and boiling points are low.

(vii) In the exceptional case of diamond, covalent bonding is very strong due to the very large cohesive energy of a solid, making this form of carbon very hard and with a high melting point.

Metallic Bonding

- Metallic bonding is the type of bonding found in metal elements. This is the electrostatic force of attraction between positively charged ions and delocalized outer electrons. The weakness of the bonding actions in a metal is due to the enlargement of the internuclear spacing.
- All valence electrons in a metal combine to form a "sea" of electrons that move freely between the atom cores. A metal may be described as a cloud of free electrons. So, metals have high electrical and thermal conductivity.
- This type of bonding is nondirectional and is rather insensitive to structure. As a result we have a high ductility of metals - the "bonds" do not "break" when atoms are rearranged – metals can experience a significant degree of plastic deformation.
- The metallic bond is weaker than the ionic and the covalent bonds.

METALLIC BONDING



The metallic bond forms when atoms give up their valence electrons, which then form an electron sea. The positively charged atom cores are bonded by mutual attraction to the negatively charged electrons

Characteristics of Metallic Compounds

- Metallic compounds are crystalline in nature
- They are good conductors of heat and electricity
- As the free electrons in a metal absorb light energy, metals are opaque to light.
- Metals have good lustre and high reflectivity
- Metals are moderately strong and generally have high melting points.

Secondary Bonds

- Secondary bonds are weaker than primary bonds. The attractive forces exist between atoms or molecules. These are also known as intermolecular bonds as they result from intermolecular or dipole attractions.
- Vander waals bonds and hydrogen bonds are common examples of secondary bonds.

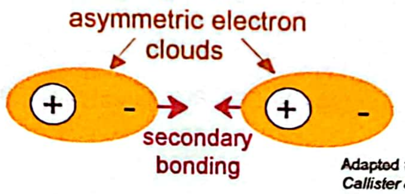
Secondary Bonding or van der Walls Bonding

- Also known as physical bonds
- Weak in comparison to primary or chemical bonds
- Exist between virtually all atoms and molecules
- Arise from atomic or molecular dipoles
  - bonding that results from the coulombic attraction between the positive end of one dipole and the negative region of an adjacent one
  - a dipole may be created or induced in an atom or molecule that is normally electrically symmetric

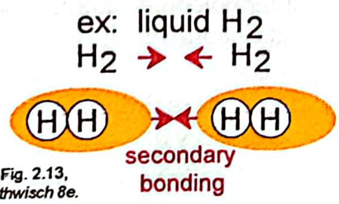
SECONDARY BONDING

Arises from interaction between dipoles

- Fluctuating dipoles



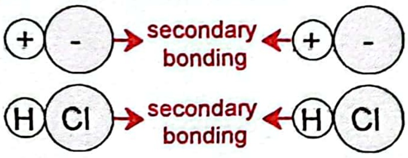
Adapted from Fig. 2.13, *Callister & Rethwisch 8e.*



- Permanent dipoles-molecule induced

-general case:

-ex: liquid HCl



Adapted from Fig. 2.15, *Callister & Rethwisch 8e.*

-ex: polymer

secondary bonding

Secondary Bonding or van der Walls Bonding

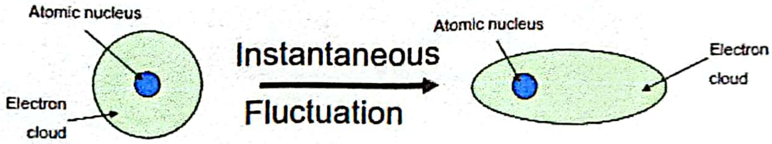
➢ Fluctuating Induced Dipole Bonds

– A dipole (whether induced or instantaneous) produces a displacement of the electron distribution of an adjacent molecule or atom and continues as a chain effect

– Liquefaction and solidification of inert gases

– Weakest Bonds

– Extremely low boiling and melting point



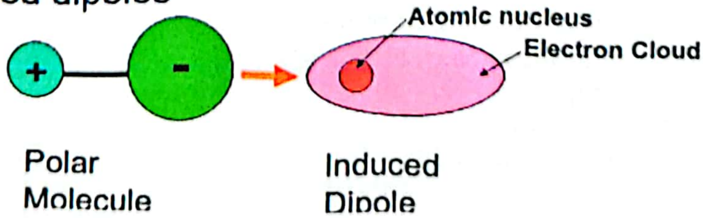
Secondary Bonding or van der Walls Bonding

➢ Polar Molecule-Induced Dipole Bonds

– Permanent dipole moments exist by virtue of an asymmetrical arrangement of positively and negatively charged regions

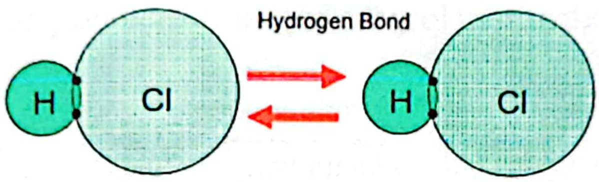
– Polar molecules can induce dipoles in adjacent nonpolar molecules

– Magnitude of bond greater than for fluctuating induced dipoles



Secondary Bonding or van der Walls Bonding

- **Permanent Dipole Bonds**
  - Stronger than any secondary bonding with induced dipoles
  - A special case of this is hydrogen bonding: exists between molecules that have hydrogen as one of the constituents



Secondary Bonding or van der Walls Bonding

- **Permanent Dipole Bonds:**
- **Hydrogen bonds in water**

Many molecules do not have a symmetric distribution/arrangement of positive and negative charges (e.g. H_2O, HCl, NH_3)